

The empirical recurrence for OEIS A235207: recovering a conjecture that is not written down

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Abstract

OEIS A235207 records “Empirical recurrence of order 62”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 2744$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 62, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 62 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A235207 is “Number of $(n+1) \times (3+1)$ 0..7 arrays with every 2×2 subblock having its diagonal sum differing from its antidiagonal sum by 5, with no adjacent elements equal (constant-stress tilted 1×1 tilings).”. It has offset 1 and begins

5696, 18444, 61160, 215996, 781040, 2971196, 11553392, 46833260, ...

The entry states:

Empirical recurrence of order 62 (see link above).

As of the “Last modified” line on the live entry (Jun 19 2022 02:22:50, revision 11) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 2744$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 2744$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 5488$ exact terms of the model returns order 62 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{62} c_i a(n-i),$$

c_1	24	c_{17}	−120330174169238	c_{33}	232954915459306475350	c_{49}	4163635305673907
c_2	−66	c_{18}	135354114065829	c_{34}	1438902327015231932393	c_{50}	4179717973400323
c_3	−2900	c_{19}	1436806566481760	c_{35}	−1122410630777447333580	c_{51}	−227282479565217
c_4	23909	c_{20}	−2884052904327915	c_{36}	−4191435142428366937230	c_{52}	−245195830193290
c_5	121626	c_{21}	−13271288781553658	c_{37}	3945009128538397107656	c_{53}	776835222980013
c_6	−2020975	c_{22}	38808327218684051	c_{38}	10181411474159613061648	c_{54}	1041662226897454
c_7	41248	c_{23}	94238218765683696	c_{39}	−10727282674729095267024	c_{55}	−10736903279509
c_8	92197459	c_{24}	−386972386607313535	c_{40}	−20646878948795301948544	c_{56}	−29017289897406
c_9	−229970590	c_{25}	−496703356612795162	c_{41}	23019068786650388928864	c_{57}	−23806963711800
c_{10}	−2648802275	c_{26}	3021747095903015047	c_{42}	34947225349274122277968	c_{58}	441006711626394
c_{11}	12126737608	c_{27}	1714251194319709624	c_{43}	−39113859798003207983552	c_{59}	106669676945630
c_{12}	49046824079	c_{28}	−18976135605382949723	c_{44}	−49259618472560886099040	c_{60}	−2183674318175
c_{13}	−366889784854	c_{29}	−1380459237052397778	c_{45}	52249589075844699682688	c_{61}	−1065465510100
c_{14}	−506390463219	c_{30}	97287014443846392831	c_{46}	57475516157400121078912	c_{62}	−1061044491386
c_{15}	7710610912720	c_{31}	−28247904409521184528	c_{47}	−53915429488715877533440		
c_{16}	−705298866691	c_{32}	−410969100562839593232	c_{48}	−54825757449387360154368		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 62, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $62 + 4$ terms removed returns order 62 as well, so $\deg q_{\min} = 62 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 19 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $62 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 62 that this sequence satisfies — and that family turns out to have exactly one member.

References

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